School of Computer Science Engineering and Information Systems
LAB ASSESSMENT TEST – II - WINTER SEMESTER-2025
Programme Name & Branch: M.Tech SE
Course Name Code: CSE3502 Course Name: Information Security Management

1. The database server in your organization's production environment is intermittently unavailable, causing downtime for the finance team. As a security analyst at TechNova, diagnose the root cause using the most effective tools and commands, and propose immediate remediation steps.
2. A critical application server is experiencing high latency and frequent crashes after a recent update. As an intern in the cybersecurity operations team at CyberGuard Inc., investigate this issue step-by-step, focusing on security-related factors like resource exhaustion or malicious activity.
3. Users report that the company VPN is not connecting during peak hours. As a network security specialist at SS Tech, perform a comprehensive diagnosis to identify potential misconfigurations or attacks, and outline the resolution process.
4. A team lead suspects that sensitive client contracts stored in a folder are being accessed by non-team members. As a security intern, apply appropriate access controls and auditing to the documents using commands, ensuring only authorized users can view them.
5. After a suspected insider threat, the integrity of log files in the audit server needs verification. As a forensic analyst at InfoSec Labs, use commands to check for tampering and implement measures to protect file integrity going forward.